

Module 2: Basic Chemistry - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. Table salt (NaCl) forms when a sodium atom gives up an electron to a chlorine atom.
What kind of bond holds the result together?
2. A. A covalent bond from shared electrons
3. B. An ionic bond between oppositely charged ions
4. C. A hydrogen bond
5. D. No bond at all
6. Answer: B
7. Why: Transferring electrons creates charged ions that attract each other, which is an ionic bond. Sharing electrons instead would make a covalent bond.
8. Why can water dissolve so many substances important to cells?
9. A. Water is polar, so its partial charges pull on charged and polar molecules.
10. B. Water carries no charge anywhere in the molecule.
11. C. Water is a pure element.
12. D. Water is always strongly acidic.
13. Answer: A
14. Why: Water's polarity (uneven charge distribution) lets it surround and separate ions and polar molecules, so they dissolve.
15. A solution changes from pH 7 to pH 4. This means it became:
16. A. more basic, with fewer hydrogen ions
17. B. exactly neutral
18. C. a pure element
19. D. more acidic, with more hydrogen ions
20. Answer: D
21. Why: Lower pH means a higher concentration of hydrogen ions, which is more acidic.

22. Blood keeps a nearly steady pH even when acids enter it. What best explains this stability?

23. A. Blood contains no water.

24. B. Acids are unable to enter blood.

25. C. A buffer resists pH change by absorbing or releasing hydrogen ions.

26. D. pH never changes in any liquid.

27. Answer: C

28. Why: Buffers stabilize pH by taking up or releasing hydrogen ions, which protects enzymes and cell function.